

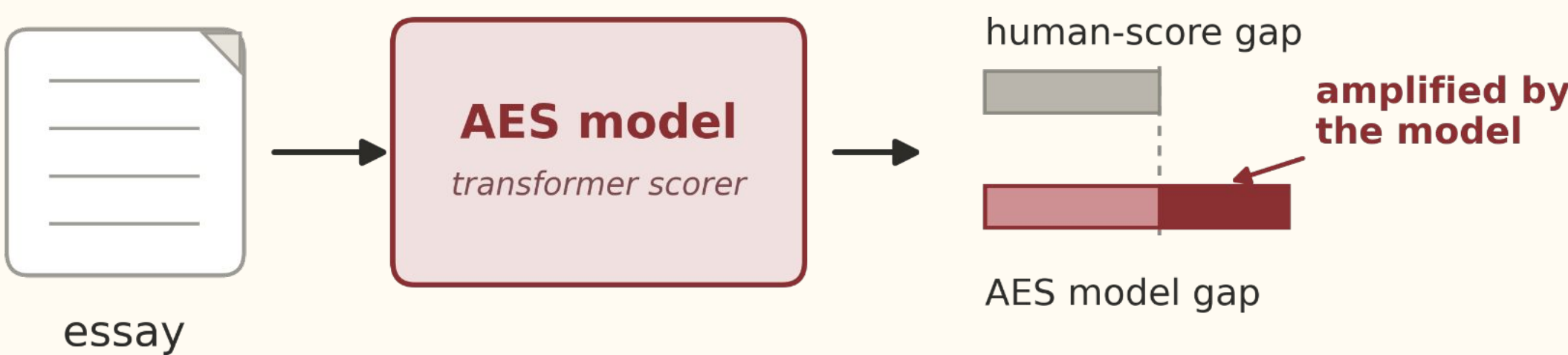


Testing Fairness Interventions in Automated Essay Scoring

When debiasing doesn't hold

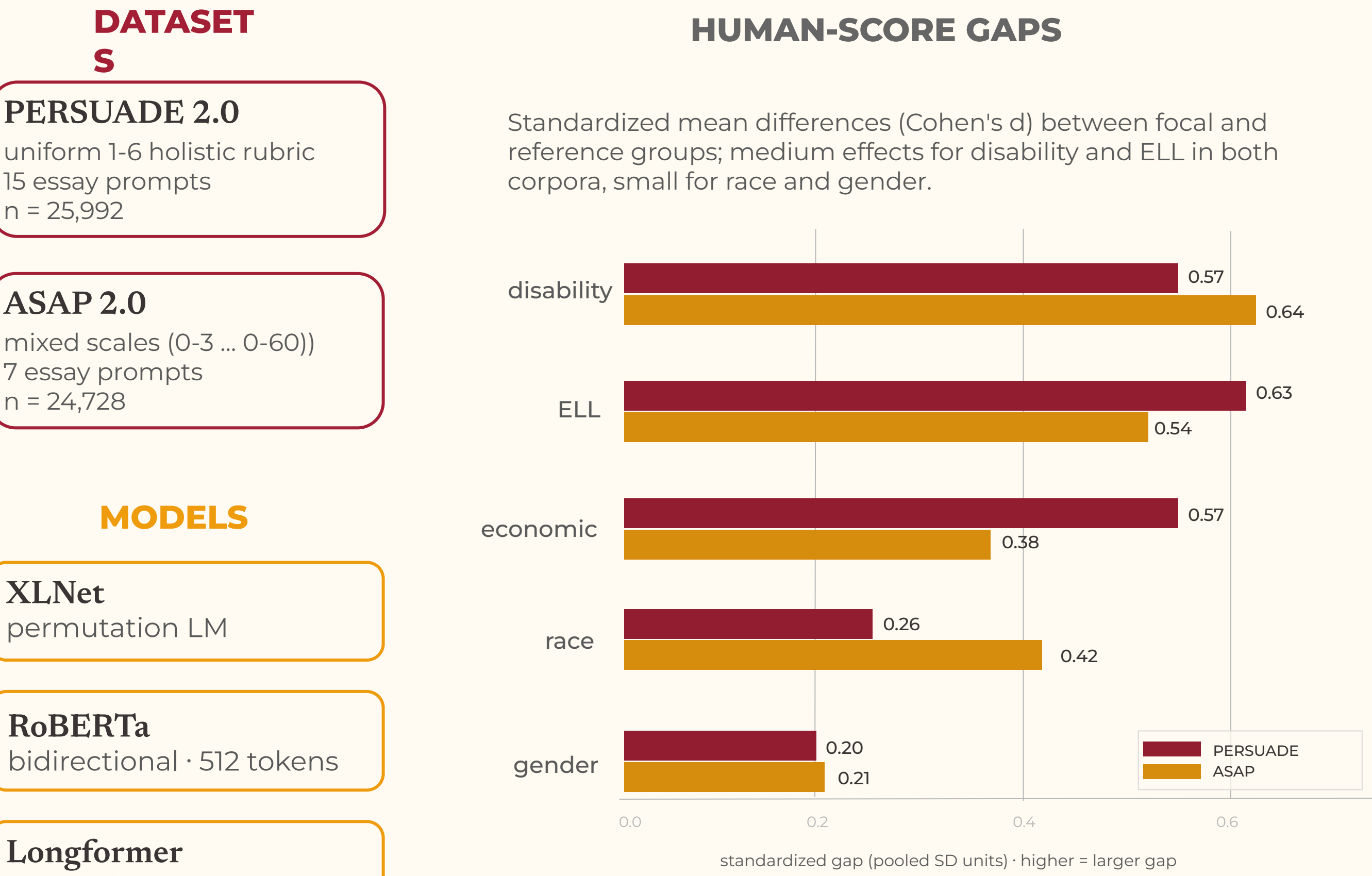
Motivation

AES runs at scale; a biased grader compounds disadvantage for already-underserved students



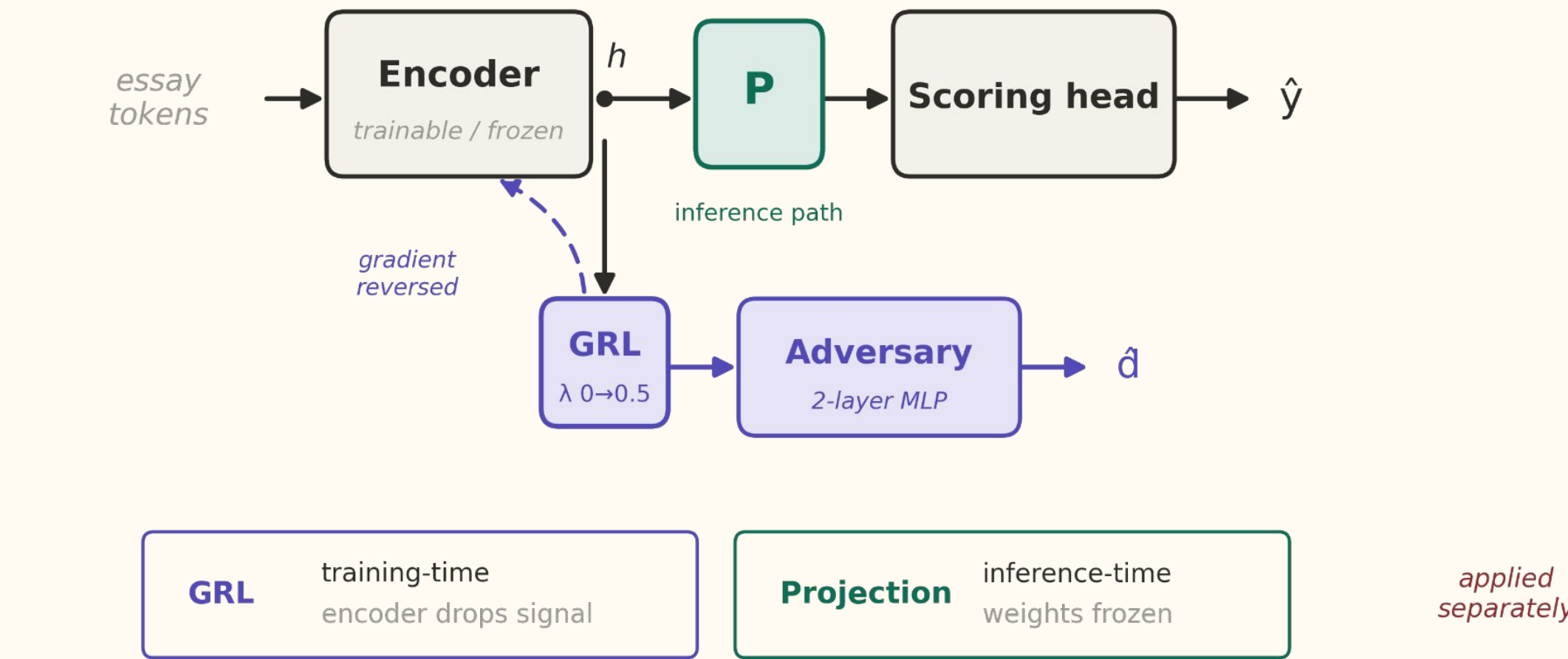
Does removing the encoded demographic signal actually debias scores?

Data and Models



Methods

Two ways to remove the demographic signal — without breaking the score

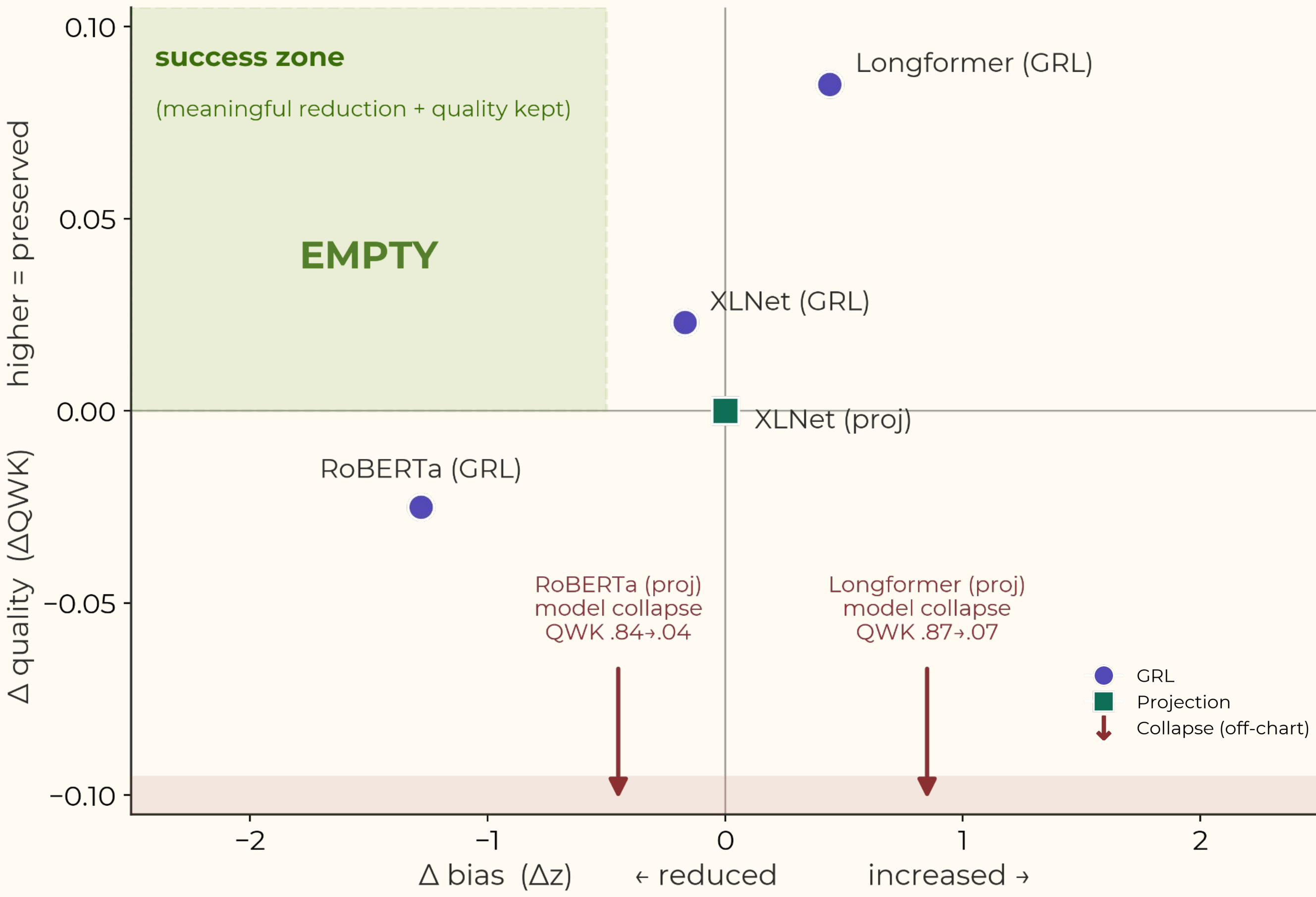


Success = bias down (z+) · signal gone (κ+) · quality kept (QWK holds)

Findings

Three failure modes, two interventions

z = does the model widen the gap · κ = is demographic info still in the model · QWK = do scores still match human raters



No method reaches the success zone — GRL trades bias for quality or backfires; projection collapses the scorer.

Metrics

BIAS	REPRESENTATION	QUALITY
Weighted regression z	Probing κ	Quadratic-weighted κ
$z = \frac{\beta}{SE_{cluster}}$	κ_{probe}	κ_w^{quad}
Does the model widen the human-score gap?	Is the demographic signal recoverable from hidden states?	Do the scores still agree with human raters?
β — per prompt group coefficient	Cohen's κ on a classifier trained to predict demographic group from the frozen feature model's hidden representation (Kwako & Ormerod, 2024)	Agreement between model and human scores, penalizing larger disagreements more (Williamson et al., 2012). The guardrail: mitigation must not crater scoring quality
$SE_{cluster}$ — standard error clustered by prompt, so effect sizes are comparable across prompts and corpora	$\kappa < 0.2$ · near-chance $\kappa \geq 0.4$ · meaningful encoding	Higher is better. We compare QWK pre → post mitigation
$z > 0$ · model <i>amplifies</i> the human gap $z < 0$ · attenuates $ z > 1.96$ · significant		
All together: z unchanged but κ lowered would mean we removed the signal in the representation without changing what the model does — the failure mode Han et al. (2021) named		

GitHub Repo



Demo



GRL Bias Table

attribute	XLNet	RoBERTa	Longformer
gender	-0.11	-0.37	-0.12
race	-0.22	-1.78	+2.00
ELL	-0.20	-2.11	-0.57
economic	+0.01	-1.55	+1.65
disability	-0.53	-0.59	+0.23
QWK	0.755→0.778	0.655→0.630	0.655→0.740

Where GRL reduces bias, it costs QWK (RoBERTa) or doesn't generalize; on Longformer it backfires — race, economic, and disability gaps widen

Mediation

Why the gap persists after debiasing

Controlling for surface features (length, sentence count, word length, Flesch-Kincaid, lexical diversity):

attribute	length only	all features	
gender	+68%	+99%	gap fully explained by writing features
ELL	-66%	-118%	gap widens when features held constant
economic	-29%	-18%	gap widens
disability	-51%	-57%	gap widens

Two failure modes:
– Gender: gap rides on essay length, a feature the scoring head retains after projection
– ELL / disability / economic: residual penalty beyond what surface features predict: distributed, non-linear, not erasable by removing one direction

Takeaways

Bias is systemic
Recurs across XLNet, RoBERTa, Longformer

The success quadrant is empty.
No model × method achieves bias down + QWK held

Representational ≠ behavioral fairness
Probe κ drops → z unchanged
Han et al. (2021) failure mode confirmed empirically

Certifying fairness on representational metrics alone is insufficient
ELL and disability gaps persist at the score level regardless

Next Steps
equal-opportunity losses, group DRO,
targeted data augmentation for ELL writers, multi-trait scoring